Pinball Machine with Arduino

By Adrien Waeles-Devaux and Redha Abderrahmane

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1) Project presentation

As part of the Arduino course in the 2nd year of the integrated preparatory program at Polytech Nice-Sophia, we chose to design a pinball machine from simple components. The main strength of this project is that it is multidisciplinary, which let us teach ourselves in fields we are passionate about, such as mechanics, electronics and programming.

2) Objective:

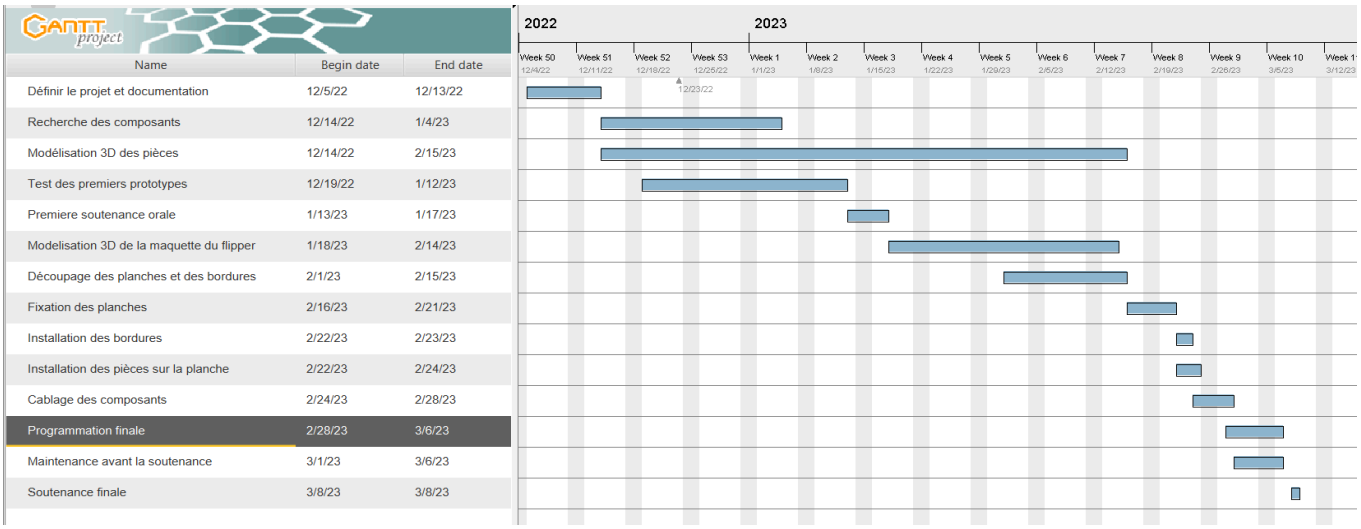
Since the rise of video games, pinball machines have become increasingly rare and seem to be disappearing. This can be explained by manufacturing and maintenance costs much higher than those of a digital game.

The “Bram Stoker’s Dracula” pinball machine and its technical complexity (Source: pinballmag.fr)

The goal of this project was therefore to show that it is possible to build your own pinball machine from simple components with Arduino, compared with machines sold new for around 10,000 euros.

3) Planning and progress:

This project required a lot of organization and commitment outside class, since the 8 sessions of 3 hours we had were far from enough to achieve our vision of the project. To make up for this lack of time, we worked every afternoon after class and during the holidays to reach the level of progress shown at the presentation. About 600 hours (adding up the work of both team members) were spent on the project up to the day of the final defense.



Name	Begin date	End date
Définir le projet et documentation	12/5/22	12/13/22
Recherche des composants	12/14/22	1/4/23
Modélisation 3D des pièces	12/14/22	2/15/23
Test des premiers prototypes	12/19/22	1/12/23
Premiere soutenance orale	1/13/23	1/17/23
Modelisation 3D de la maquette du flipper	1/18/23	2/14/23
Découpage des planches et des bordures	2/1/23	2/15/23
Fixation des planches	2/16/23	2/21/23
Installation des bordures	2/22/23	2/23/23
Installation des pièces sur la planche	2/22/23	2/24/23
Cablage des composants	2/24/23	2/28/23
Programmation finale	2/28/23	3/6/23
Maintenance avant la soutenance	3/1/23	3/6/23
Soutenance finale	3/8/23	3/8/23

The Gantt chart above shows the progress of the project over our 13 weeks of work. 3D modeling took a considerable amount of time, since we designed every part of our pinball machine. Manufacturing and

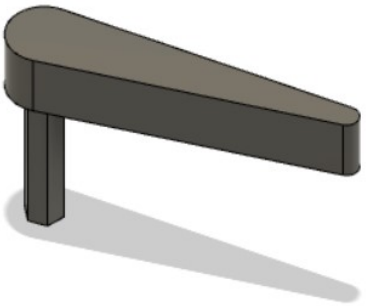
assembly also took a large part of our time, since it was the first time we worked on a project of this scale, so we lacked experience in making and assembling technical parts.
Learning new software and new tools took significant time before reaching a good level of practical and theoretical mastery and getting the best possible result. This learning time explains the differences between our initial plan, available on our GitHub, and this chart. They also come from the delivery time of the components, which required us to reorganize so as not to lose time.

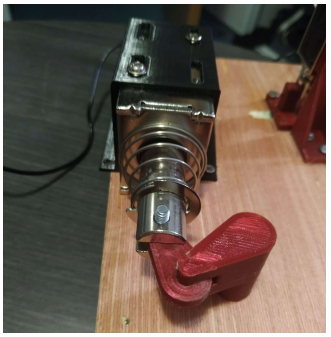
4) Presentation of the elements of our pinball machine:

Online documentation on how pinball machines work is quite rich and very instructive; however, the mechanics of a pinball machine are so complex that it was not possible to use the same components and mechanisms while keeping the project affordable and delivering it on time.
So we had to design all our elements ourselves.
To do this, we first built prototypes on small wooden boards so that we could run many tests before the final version. In total, more than 25 parts were modeled in 3D, each one requiring many changes and tests.

Here are the different elements we built:

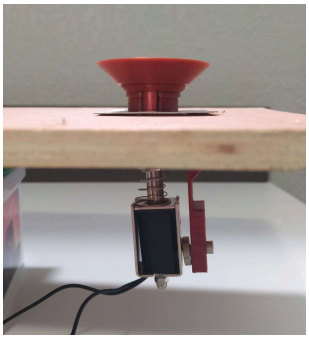
The flippers:
The parts that give the game its French name (“flipper”), controlled by the player to hit the ball and send it to the different areas of the game.





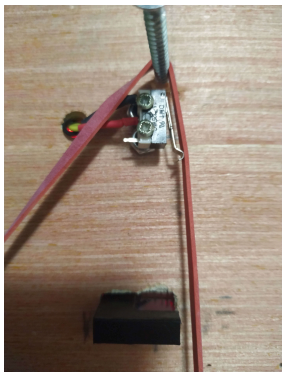
They are driven by 2 powerful electromagnets through a mechanism, modeled and 3D-printed, that converts their linear motion into a rotation.

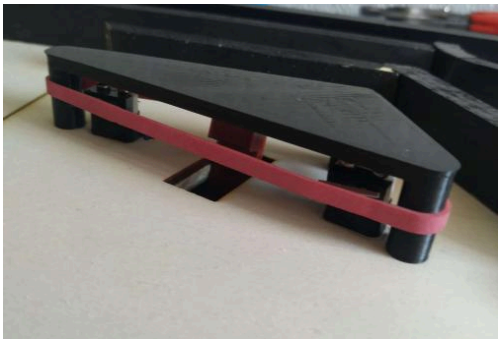
The bumpers:



Iconic mushroom-shaped parts that come down on the ball when it touches them, to send it back in the opposite direction. They are often arranged in a triangle, close to each other, so that the ball bounces back and forth between them to score as many points as possible.

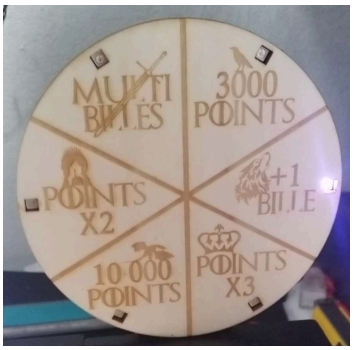
The bumpers are triggered when the ball touches both a copper wire inserted in the bumper and a metal plate set into the board, as detailed in the wiring section.
The slingshots:
Parts fitted with a rubber band that sends the ball back in the opposite direction to put it back in a playable area. They are triggered by 2 microswitches placed at the ends of each rubber band, which detect the impact of the ball.





The targets:
They also work with microswitches. When the ball hits a target, it presses its switch. The signal received by the Arduino is used to count the points.

The wheel of fortune:



An element unique to our pinball machine. When it is triggered (for example after the player has hit all the targets), it awards a randomly chosen bonus. It works with a light animation of NeoPixels that light up one after the other in a circle; the animation ends with one NeoPixel blinking to show the bonus won on the wheel.

The launch handle:

The handle is fully 3D-printed. So that it can be inserted into a holder in which it slides, and so that a spring can be fitted on it, it was modeled in 3 parts. This also limits the supports needed for 3D printing. The 3 parts screw together thanks to a thread made in Fusion 360. The photo opposite shows the 3 assembled 3D-printed parts, with the spring that launches the ball.

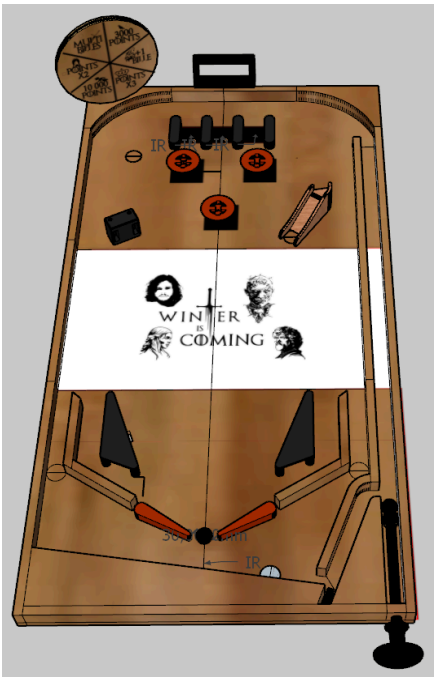
Launch area:

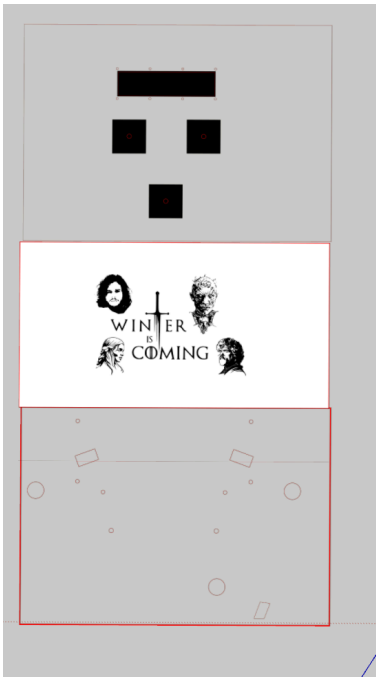
When the player fails to catch a ball, it must be brought back to the launch area so that the player can have another try. For this we designed a small plastic part attached to a solenoid which, when activated, pushes the ball just above the handle.

The ramp:

The entrance of our ramp is fully modeled and 3D-printed; for the rest of the ramp, we soldered several stiff copper wires together, keeping a constant spacing along the whole ramp so that the ball does not fall.

5) Project model:

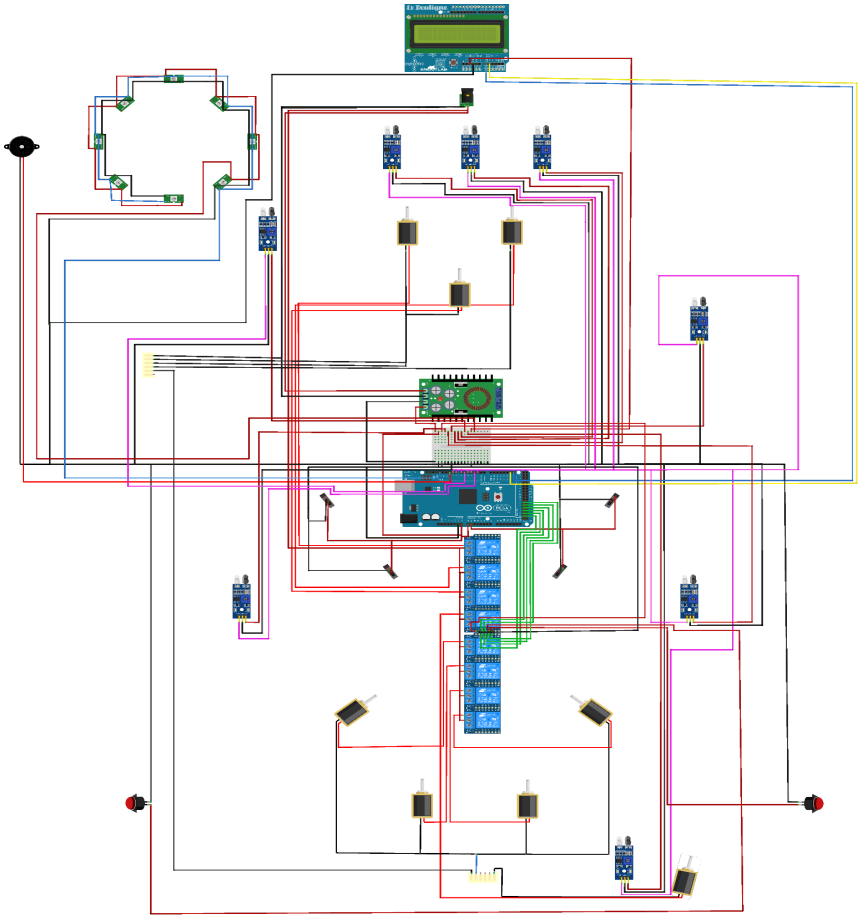




Before assembly, we decided to build a complete model of the project by importing and placing all the elements modeled during the design phase, to avoid many placement errors.

It was very useful: it let us cut our boards with the holes for fixing and positioning our components already in place, which saved a considerable amount of time. It was also the basis for designing the borders of the pinball machine, drawn in the same software and then converted into 2D vector files so they could be laser cut.

6) Wiring diagram:

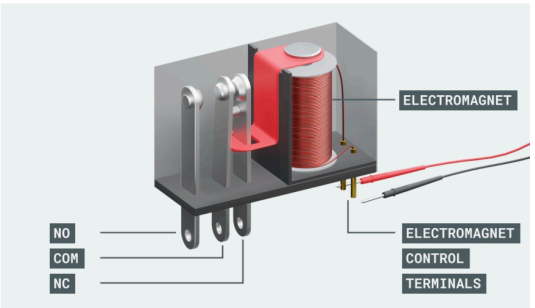


Wiring diagram of the project in Fritzing

First, here are a few notions to better understand the details of the wiring:

- Solenoids are electromagnets that convert electricity into mechanical motion by generating a magnetic field. Inside, a wire is wound into a coil. When current flows through the wire, it generates a magnetic field, which pulls the metal plunger inwards. When the current is cut, the solenoid's spring brings the plunger back to its initial position.

- Relays are components used to control external electrical loads from digital signals. When the signal is applied, the relay switches a contact with an electromagnet to open or close an electrical circuit.
  In our case, the relays isolate the control circuit (driven by the Arduino, which works at around 5V) from the power supply of the solenoids, which works at 24V.
  The diode built into the relay modules only protects the relay's coil: each solenoid needs its own diode, connected in parallel across its terminals, to absorb the voltage spike produced when the current is cut.



In our project, we needed solenoids able to send the ball very fast and responsively to every playable area of our pinball machine. So we chose solenoids with a force of 65N running at 24V.
This requirement forced us to power our project with a power supply able to deliver 24V and 2A.

The solenoids are connected in parallel so that they can be activated individually or at the same time, and their grounds are all connected to a common ground.

The two flipper solenoids are controlled directly by the player with push buttons mounted on the borders of the board. These switches are connected directly to the relays, so that the solenoids are activated as soon as the circuit is closed.

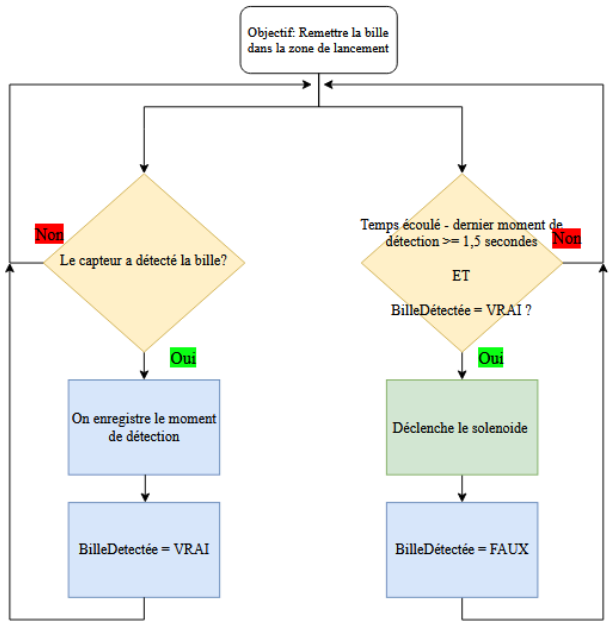
To power the brain of our project, the Arduino Mega board, we used a step-down voltage converter that takes 24V as input and outputs 5V.

It also powers the following low-voltage components:
- Microswitches
- LCD screen
- Buzzer
- Infrared sensors
- NeoPixels

Since the diagram is already very busy, we did not show the wiring of the bumpers' metal plates and rings.
It works simply: the metal rings are connected to Arduino inputs, held at 5V by the internal pull-up resistor, while the metal plates are connected to the common ground. Since the ball is made of metal, when it touches both the ring and the plate, the circuit closes and the bumper is triggered.

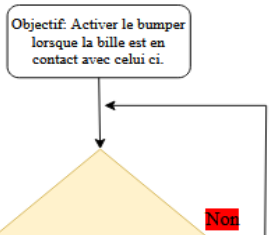
7) Algorithm:

Although fairly simple, the algorithm is quite long to explain in detail. Our explanations therefore cover only the most important points, and the commented code is available on GitHub.
Let us detail how the launch area works:



The goal is to bring the ball back above the launch handle when the player loses it.
An infrared sensor detects the moment the ball leaves the game. We then wait a moment for the ball to settle in front of the solenoid. Activating the solenoid then pushes the ball along a path bounded by our borders up to its launch position.
Our challenge was to avoid any blocking pause between 2 commands, so as not to suspend the whole program.

Here is how it works: if the ball passes over the IR sensor, we record the time of detection in milliseconds and a state variable “ball detected” is set to “true”. The program continuously checks whether the time elapsed since this detection is greater than



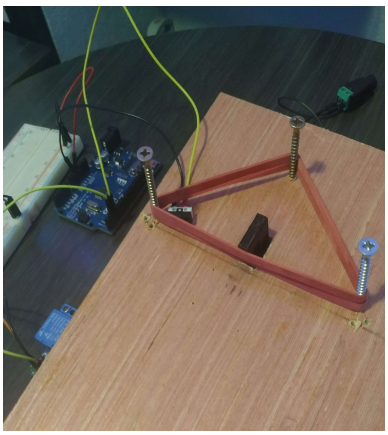
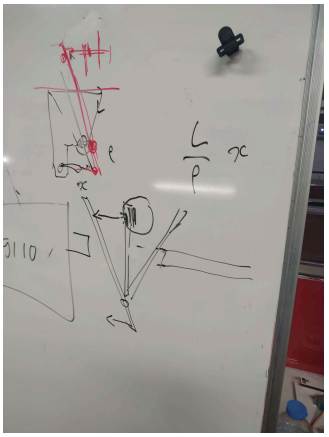
1.5 seconds and whether “ball detected” is true. If so, the solenoid is triggered. Finally, “ball detected” is set back to “false” so that the algorithm can run again.

The bumpers work in a much simpler way:
The Arduino pin connected to the bumper's ring is set to INPUT_PULLUP to use the Arduino's built-in pull-up resistor.
We then want to know whether the ring/metal plate circuit is closed. To do this, we continuously check whether the ring is at zero volts (that is, connected to ground). If so, we activate the solenoid and increase the score.

To activate the solenoid for 100 milliseconds, we can simply use a delay, because it is so short that the ball will not have time to trigger another element in the meantime. Also, the flippers are not controlled by the Arduino, as explained in the wiring section, so this does not disturb the game.

8) Difficulties encountered:

Since we had to go through every stage of a project, from design to assembly, we ran into several difficulties:

- First, our first slingshot prototype could not send the ball back hard enough with the hardware we had. It was made of 3 screws stretching a rubber band and a small 3D-printed arm pushing it (see the first photo above).
To solve this, we designed a new system based on a lever, which gives a larger range of motion and more force.
We modeled a first 3D part on which we tested different pivot points and lengths. Then we tested this prototype on a board and modified the part based on these results to make and print the final version (see the photo above on the right).

- The ball was not caught in the launch area because the solenoid pushed it too fast. The solution was to adjust the solenoid's activation time in the Arduino program, which sends the ball with more or less force.

- The bumpers were not sensitive enough: the ball did not always make contact between the ring and the metal plate. So we adjusted their height so that the ball can touch the bottom of the bumper.

- Finally, the drawback of the infrared sensors in our project is that they depend on the lighting of the room. To improve the project, we would like to replace them with a more reliable type of sensor.

9) Project cost

Estimated material cost of the project (based on Amazon prices):

Electronic components:

- Arduino Mega board = €25
- 2 solenoids 24V 65 newtons = €60
- 6 solenoids 24V 8 newtons = €60
- Step-down converter = €8
- 8 relays = €14
- 8 infrared sensors = €10

16x2 character LCD + I2C module = €10
6 NeoPixels = €5
2 push buttons = €6
6 microswitches = €6
24V 2A power supply = €15

Miscellaneous parts:

- 3 balls, 16 mm = €5
- Large rubber bands = €2
- Screws and bolts ≈ €15

Boards and 3D filament:

3D-printed parts (PLA): 300g ≈ €15
4 wooden boards, 3 mm thick = €9

Total cost: about €265

10) Future work

This project had to be completed in 13 weeks to meet the deadline, which forced us to focus on the essentials so that we could present a first finished version on time. However, there are many possible improvements. We would like to add NeoPixel strips to create interactive lighting that changes color or blinks depending on the game elements hit, or add new elements for the player to aim at.
We started designing a multiball mode with a 3D-printed part and a stepper motor that releases several balls at once.
Finally, we want to keep decorating and theming the pinball machine with 3D printing, laser engraving and custom stickers.

11) Conclusion:

We were passionate about this project: the complexity of its mechanics and the variety of electronic components taught us a lot. We developed new working methods for research and development and became more efficient and autonomous.
This project also confirmed our study choices: Redha Abderrahmane wants to specialize in Electronics and Adrien Waeles-Devaux in Robotics.

To help make building a pinball machine more accessible, all the documentation, 3D models and other manufacturing files are available as open source on this GitHub repository.

12) Bibliography:

Documentation:

Flipper (pinball), Wikipedia
wikipedia.org/wiki/Flipper
Pinball Makers (resources for building your own pinball machine)
pinballmakers.com

Videos:

“How a Pinball Machine works in Slow Motion - The Slow Mo Guys”: youtu.be/Tmg5WOvPKpU
“Wooden Pinball + Arduino DIY”: youtu.be/y2FeiB1704w
“Fabriquer un FLIPPER, un ENFER” (in French): youtu.be/RLwirZu9_jc
Inspiration for how the bumpers work: youtu.be/yfZipK-Nqeo
“DIY 3D printed pinball machine”: youtu.be/SnD3WfmEfXY

3D files used:

Bumper, completely modified to fit our project: www.thingiverse.com/thing:3286052
Shims to align the flippers at the right angle:
www.thingiverse.com/thing:1644154